Approximate HJ Reachability via Importance Sampling

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Abstract

Computing safe reachable sets in verification settings, via viscous Hamilton-Jacobi (HJ) partial differential equations, involves a grid discretization of the phase space for a numerically consistent optimal solution. This presents a two-fold problem: discretization implies exponential memory requirements; and, computational costs increase as problem dimensions multiply. While parallel computing can accelerate on GPUs and reduce the time to compute solutions, it does not address the exponential memory costs associated with the grid discretization. Linearizing the HJ PDE, we extend the solutions to the original viscous problem to those of the reachable backward reach tubes and reach-avoid tubes and we cast the recovery of the safe set as the integration of a Gaussian process that approximates the HJ variational PDE from the recovered reachable set. This paper presents a storage-free means for resolving safe sets in a computationally time-efficient manner without dealing with the exponential memory costs of grid discretization.

1. Introduction

We explore the numerical computation of safe reachable sets within the backdrop of Hamilton-Jacobi (HJ) equations. Within optimal control contexts, enforcing consistency of the numerical solution to the underlying partial differential equation is typically carried out via spatial discretization on a grid whilst integrating the PDE with total-variation diminishing Runge-Kutta algorithms (Crandall & Majda, 1980b;a; Crandall & Lions, 1984; Osher & Shu, 1991). However, this incurs a quadratic computational cost on the underlying problem such that solutions are only possible in finite dimensions (typically less than 6-10 spatial dimensions on a single workstation). Recent efforts explored accelerating the numerical solution to HJ-based safe reachable sets by exploring grid discretization on accelerated graphical processing units (GPUs) (Molu, 2025; 2024a) with reports of increased computational speeds on moderately high-dimensional problems. However, such methods are limited by the availability of large computing resources and do not address algorithmic improvement of the computational scheme. This paper takes a further step by minimizing the computational cost for solving time-dependent Hamilton-Jacobi (HJ) partial differential equations (PDEs) (Evans, 2022). We work within viscous HJ reachability ramification (Lygeros, 2004; Mitchell et al., 2005; Molu, 2024b; 2025) and establish our results with globally optimal solutions to reachability problems in nonlinear settings.

We consider first-order nonlinear scalar HJ PDEs of the form

$$\boldsymbol{v}_t + \boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{v}) = 0 \text{ in } \Omega \times (0, T], \\ \boldsymbol{v}(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}) \text{ on } \partial\Omega \times \{t = 0\} \qquad \text{(HJ-IVP)}$$

where the state $\boldsymbol{x}$ belongs to the open set $\Omega \subseteq \mathbb{R}^n$, the boundary of Ω is denoted $\partial\Omega$; $\boldsymbol{v}_t$ denotes the partial derivative of the vector field $\boldsymbol{v}(t; \boldsymbol{x})$ with respect to time, t; the Hamiltonian $\boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{v})$ is continuously defined on $\mathbb{R}^n$, where $D\boldsymbol{v}$ is the spatial gradient of $\boldsymbol{v}(t; \boldsymbol{x})$; and $\boldsymbol{g}(\boldsymbol{x})$ is bounded and uniformly continuous (BUC) in $\mathbb{R}^n$.

While a unique C^2 solution $\boldsymbol{v}(t; \boldsymbol{x})$ to (HJ-IVP) exists uniformly on $0 \leq t < T$ for which $\lim_{t \to T} \boldsymbol{v}(t; \boldsymbol{x})$ (when $\boldsymbol{H}$ and $\boldsymbol{g}$ are smooth and $\boldsymbol{v}_0(t; \boldsymbol{x})$ is compactly supported), $\boldsymbol{H}(t; \cdot)$ lacks continuous derivatives in this limit so that $D\boldsymbol{v}$ *vanishes* at $t = T$. However, one may find Lipschitz continuous functions $\boldsymbol{v}(t; \boldsymbol{x})$ on $\mathbb{R}^n \times (0, T]$ which are smooth *almost everywhere* (a.e.) and satisfy (HJ-IVP) in a "generalized sense". Suppose that we fix a viscosity term $\delta > 0$, Crandall and Lions (Crandall & Lions, 1984) showed that $\boldsymbol{v}^\delta$ is the *vanishing viscosity* solution of

$$\boldsymbol{v}_t^\delta + \boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{v}^\delta) = \frac{\delta}{2} \Delta \boldsymbol{v}^\delta \text{ in } \mathbb{R}^n \times (0, T]; \\ \boldsymbol{v}^\delta(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}) \text{ on } \mathbb{R}^n \times \{t = 0\} \qquad \text{(HJ-Visc)}$$

with Laplacian $\Delta \boldsymbol{v}^\delta$. For a constant $k > 0$, uniform convergence of $\boldsymbol{v}^\delta \to \boldsymbol{v}$ exists in the limit $\delta \to 0^+$, i.e.

$$\sup_{t \in (0,T]} \sup_{x \in \mathbb{R}^n} | \boldsymbol{v}(t; \boldsymbol{x}) - \boldsymbol{v}^\delta(t; \boldsymbol{x}) | \leq k\sqrt{\delta}. \qquad (3)$$

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In $\inf\text{-}\sup$ or $\sup\text{-}\inf$ optimal control problems (Lygeros, 2004), the Hamiltonian of (HJ-Visc) is related to the *backward* reachable set (Mitchell et al., 2005) of a dynamical system. Reachable systems are those where one can compute all states that can be reached from an initial condition in *a finite number of steps*. Popular numerically consistent optimal solutions (Crandall & Lions, 1984; Crandall & Majda, 1980a) to (HJ-Visc) utilize problem space discretization (Osher & Fedkiw, 2004; Mitchell et al., 2005) on a grid. However, this exacts an exponential memory storage — making their practical applications on large problems limited.

Recent efforts (Molu, 2024b; 2025) leveraged single-instruction, multiple threads (SIMT) with modern GPUs to accelerate computation on a host of practical problems — showing improved spatial scalability with up to $76 - 92\%$ computational improvement on higher-dimensional problems. However, this employed grid-storage of the value function; as problem dimensions increase, the need for more memory renders such approaches unattractive.

Despite the success of the viscous solution (HJ-Visc), the structure of the possible singularities of the solution $\boldsymbol{v}$ and the convergence details of $\boldsymbol{v}^\delta \to \boldsymbol{v}$ leaves open questions: even though (HJ-IVP) holds a.e., it is not clear that $D\boldsymbol{v}^\delta \to D\boldsymbol{v}$ a.e (Evans, 2010). This paper will use new understanding relating to the linearization of (HJ-Visc) in improving the numerical solutions to the safe set that results from the variational version of (HJ-Visc).

The rest of this work is structured as follows: section 2, sets the background for this work. Section 3 describes the new machinery.The paper is concluded in section 5.

2. Background and Preliminaries

We first introduce the notations that are commonly used throughout the rest of this article. Reachable sets within the context of two person games (Isaacs, 1999) and their accompanying "viscous" terminal HJ PDE (Evans & Souganidis, 1984a) are then introduced. This is followed by the HJ-Isaacs (HJI) PDEs commonly used to characterize reachable sets. A complete background that sets this work in context is provided in Appendix A.

2.1. Notations and Terminologies

Conventions: Upper-case and lower-case bold font Roman letters are matrices and vectors, respectively; calligraphic letters are sets. Time variables e.g. t_0, t, τ, T will always be real positive numbers. The n-dimensional Euclidean space is $\mathbb{R}^n$; the set of natural numbers is $\mathbb{N}$. The natural log of T is denoted $\log T$. The state $\boldsymbol{x}$ belongs to the open set $\Omega \subseteq \mathbb{R}^n$. The HJ value function has a terminal value defined on the boundary of Ω, denoted $\partial\Omega$. The closure of Ω is $\bar{\Omega}$. The dot product is signified by $\langle \cdot, \cdot \rangle$. We will use D to denote the gradient of a function $\boldsymbol{x}$ of n variables and $\nabla \boldsymbol{w} = (\boldsymbol{w}_{x_1}, \cdots, \boldsymbol{w}_{x_n}, \boldsymbol{w}_t) := (D\boldsymbol{w}, \boldsymbol{w}_t)$ as the full gradient of a function of $n+1$ variables, $\boldsymbol{w} = \boldsymbol{w}(\boldsymbol{x}, t)$. Where clarity is necessary, we may write $D_y w$ to mean the spatial derivative of w with respect to the vector $\boldsymbol{y}$.

The unique solution $\xi \in \mathbb{R}^n$ of the dynamical system $\dot{\boldsymbol{x}}(\tau) = f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w})$, influenced by a pursuing player $\boldsymbol{P}$ (with control $\boldsymbol{w} \in \mathcal{W} \subseteq \mathbb{R}^p$) and its evading pair $\boldsymbol{E}$ (with control $\boldsymbol{u} \in \mathcal{U} \subseteq \mathbb{R}^m$), results in system trajectories $\xi_{\boldsymbol{x},\tau}^{\boldsymbol{u},\boldsymbol{w}}$ that are obtained by optimizing a scalar-valued payoff functional, $\boldsymbol{\ell}(\xi_{\boldsymbol{x},t}^{\boldsymbol{u},\boldsymbol{w}}(\tau))$ i.e.

$$\mathcal{J}(\tau; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) = \min_{\tau \in [t,T]} \boldsymbol{\ell}(\xi_{\boldsymbol{x},t}^{\boldsymbol{u},\boldsymbol{w}}(\tau)) \quad \text{(Payoff function)}$$

within a two-person differential game for a fixed $T \geq \tau > t$, where $\boldsymbol{\ell}$ can be a signed distance to the (boundary of the) target set/region, for example. The controllers belong in compact sets which are measurable functions i.e. $\bar{\mathcal{U}} \equiv \boldsymbol{u} : [t, T] \to \mathcal{U}$, $\bar{\mathcal{W}} \equiv \boldsymbol{w} : [t, T] \to \mathcal{W}$.

2.2. A Differential Game with Two Inputs

This section provides a cursory background that is suitable for a minimal problem contruction. Readers interested in a more detailed background can consult the thorough treatment in appendix A.

For a (possibly infinite) *terminal time* T, a constant k, controls $\boldsymbol{u} \in \bar{\mathcal{U}}$, $\boldsymbol{w} \in \bar{\mathcal{W}}$ and all $\hat{\boldsymbol{x}}$, $\boldsymbol{x} \in \mathbb{R}^n$, let the system possess a bounded and uniformly continuous (BUC) payoff functional $\boldsymbol{g}(\boldsymbol{x}(T))$ that satisfies $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}$. The differential game's *lower value* (Evans & Souganidis, 1984b),

$$\boldsymbol{v}(t; \boldsymbol{x}) = \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \mathcal{J}(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}), \tag{4a}$$

$$\triangleq \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \min_{\tau \in [t,T]} \boldsymbol{\ell}(\xi_{\boldsymbol{x},t}^{\boldsymbol{u},\boldsymbol{w}}(\tau)) \tag{4b}$$

is the viscosity solution to (HJ-IVP) (Evans & Souganidis, 1984b, Lemma 2.1) with Hamiltonian,

$$\boldsymbol{H}(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}, p) = \max_{\boldsymbol{u} \in \mathcal{U}} \min_{\boldsymbol{w} \in \mathcal{W}} \langle f(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}), p \rangle. \tag{5}$$

Here, p is the co-state and the pursuer $\boldsymbol{P}$ possesses a non-anticipative strategy $\mathcal{B} : \bar{\mathcal{U}}(t) \to \bar{\mathcal{W}}(t)$, beginning at time t, for each $t \leq \tau \leq T$ and $\boldsymbol{w}, \hat{\boldsymbol{w}} \in \mathcal{W}(t)$ such that $\boldsymbol{w}(\tau) = \hat{\boldsymbol{w}}(\tau)$ for $t \leq \tau < T$ almost everywhere. This implies that $\mathcal{B}[\boldsymbol{u}](\tau) = \mathcal{B}[\hat{\boldsymbol{u}}](\tau)$ for $t \leq \tau < T$ almost everywhere (Evans & Souganidis, 1984b).

2.3. Robustly Controlled Backward Reachable Set and Tube

Suppose that the goal of $\boldsymbol{E}$ is to reach a user-specified target region $\mathcal{L}_0(\tau)$ within $\tau \leq T$ time steps of playing the game; while $\boldsymbol{P}$ simultaneously seeks to prevent this from happening. The invariant set $\mathcal{L}_0$ obtained at T, i.e.

$$\mathcal{L}_0(T) = \{\boldsymbol{x} \in \mathbb{R}^n \,|\, \boldsymbol{v}(0;\boldsymbol{x}) \leq 0\} \tag{Target-Set}$$

is "*robustly controlled*" for the "*distance-to-target-set*" cost $\boldsymbol{g}(0;\boldsymbol{x})$, with constraints $|\boldsymbol{g}(0;\boldsymbol{x})| \leq k$, $|\boldsymbol{v}(0;\boldsymbol{x}) - \boldsymbol{v}(t;\hat{\boldsymbol{x}})\ | \leq k|\boldsymbol{x} - \hat{\boldsymbol{x}}|$ where $k > 0$ is a constant, and all $-T \leq t \leq 0$, $\{\hat{\boldsymbol{x}},\, \boldsymbol{x} \in \mathbb{R}^n\}$[1]. The distance to $\mathcal{L}_0(\tau)$ is typically found by optimizing $\boldsymbol{\ell}(\boldsymbol{x}, t)$ as in (Payoff function). The HJI PDE (4a) and Hamiltonian (5) for this set admit the form

$$\begin{aligned} &\boldsymbol{v}_t(t;\boldsymbol{x}) + \min\{0, \boldsymbol{H}\left(t;\boldsymbol{x}, D\boldsymbol{v}(t;\boldsymbol{x})\right)\} = 0, && (6)\\ &\boldsymbol{v}(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}). && \text{(HJI-RCBRT)} \end{aligned}$$

For the safety problem setup in (4a), the corresponding *robustly controlled backward reachable* ***tube*** (RCBRT) (Mitchell, 2020) on $(0, T]$ is the closure of the open set,

$$\begin{aligned} \mathcal{L}([\tau, 0],\mathcal{L}_0) = \{&\boldsymbol{x} \in \mathbb{R}^n \,|\, \exists\, \beta \in \bar{\mathcal{W}}(t) \forall\, \boldsymbol{u} \in \mathcal{U}(t), && (7)\\ &\exists\, \tau \in [-T, 0], \boldsymbol{\xi}(\bar{t}) \in \mathcal{L}_0\}. && \text{(Target-Tube)} \end{aligned}$$

Read: The set of states from which the strategies of $\boldsymbol{P}$ and for all controls of $\boldsymbol{E}$ imply that we *reach and remain in the target set* in the interval $[-T, 0]$. Following Lemma 2 of (Mitchell et al., 2005), the states in the reachable set admit the following properties w.r.t the value function $\boldsymbol{v}$,

$$\begin{aligned} &\boldsymbol{x}(t) \in \mathcal{L}(\cdot) \implies \boldsymbol{v}(t;\boldsymbol{x}) \leq 0, && (8a)\\ &\boldsymbol{v}(t;\boldsymbol{x}) \leq 0 \implies \boldsymbol{x}(t) \in \mathcal{L}(\cdot). && \text{(Zero Levelset)} \end{aligned}$$

In *backward reach avoid tubes*, the agent must avoid the unsafe region at all times. We can write (HJI-RCBRT) a *robustly controlled backward reach-avoid tube* (RCBRAT) as,

$$\begin{aligned} &\min\{\boldsymbol{v}_t(t;\boldsymbol{x}) + \boldsymbol{H}\left(t;\boldsymbol{x}, D\boldsymbol{v}\right), \boldsymbol{g}(t;\boldsymbol{x}) - \boldsymbol{\ell}(t;\boldsymbol{x})\} \leq 0,\\ &\boldsymbol{v}(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}). && \text{(HJI-RCBRAT)} \end{aligned}$$

Challenges in optimizing the RCBRT/RCBRAT: Computing the RCBRT presents a host of challenges for optimization because (a) the dynamics $f(\cdot)$ and the nonlinear PDE are usually nonlinear with coupled states in most problems of interest; (b) although $\mathcal{L}([\tau, 0], \mathcal{L}_0)$ is mostly smooth, it possesses discontinuous "shocks" at solution crossover points (see (Mitchell, 2020)); this makes global gradient-based optimization impossible for resolving $\boldsymbol{v}$; (c) more than one trajectory can emanate from the RCBRT's boundary that end up at the same point on the target set – this makes sampling-based optimization difficult, but not insurmountable; and (d) the RCBRT is nonconvex.

To mitigate these challenges above, we introduce a set of tools that allow us to circumvent these challenges.

[1]Time is reversed in BRT computational scenarios.

3. Approximate Solutions to HJ PDEs

We now transform the nonlinear hyperbolic "smoothed" HJ PDE (HJ-Visc) to a linearized form and extract its unique solution.

3.1. HJ Linearization

Here, we first transform the smooth HJ equation (HJ-Visc) into a linear form. When convenient we will drop the arguments of $\boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta)$ for ease of readability so that $\boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta) := \boldsymbol{H}^\delta$. We set

$$\boldsymbol{c}(t;\boldsymbol{x}) = \frac{2}{\delta} \cdot \boldsymbol{H}^\delta / |D\boldsymbol{v}^\delta|^2 \tag{9}$$

and let $\boldsymbol{c}(t;\boldsymbol{x})$ be locally frozen as a spatially-varying coefficient rather than a functional of $\boldsymbol{v}^\delta$; and for the terminal value $\boldsymbol{g}(0;\boldsymbol{x})$ we write $\boldsymbol{g}(\boldsymbol{x})$. The transformation that linearizes (HJ-Visc) can be written as (derivation in Appendix B),

$$\boldsymbol{\omega}^\delta := \exp(-\boldsymbol{c}\boldsymbol{v}^\delta). \tag{Linear-Trans}$$

Applying the transformation to (HJ-Visc), the fundamental solution to the equation associated with the smoothed version of the Hamilton-Jacobi equation i.e. (HJ-Visc) is delineated in Proposition 3.1.

Proposition 3.1. *Associated with (HJ-Visc) is the equation*

$$\begin{aligned} &\boldsymbol{\omega}_t^\delta - \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta = 0 \text{ in } \quad \Omega \times (0, T]\\ &\boldsymbol{\omega}^\delta = \exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right) \text{ on } \partial\Omega \times \{t = 0\}. && \text{(Linear-HJ)} \end{aligned}$$

The solution to (Linear-HJ) admits the following unique and explicit (Green's convolution) representation

$$\begin{aligned} \boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = \tfrac{1}{(\sqrt{2\pi\delta t})^n} \int_\Omega \exp\left(-\frac{1}{2\delta t} \mid \boldsymbol{x} - \boldsymbol{y} \mid^2\right) \cdot\\ \exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right) d\boldsymbol{y}, \quad ,(\boldsymbol{x} \in \Omega,\ t > 0), && (10) \end{aligned}$$

which fundamentally changes the PDE from a trivial ODE to the heat equation. In addition, (10) can be written as the expectation of the following Gaussian density

$$\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right) d\boldsymbol{y}\right]. \tag{11}$$

Lemma 3.2 (Smoothed HJ Solution)**.** *The explicit solution to the smoothed Hamilton-Jacobi equation (HJ-Visc) is*

$$\begin{aligned} \boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}(t;\boldsymbol{x}) \log\left\{\frac{1}{(\sqrt{2\pi\delta t})^n} \int_\Omega \exp\left(-\frac{1}{2\delta t}\cdot\right.\right.\\ \left.\left. |\boldsymbol{x} - \boldsymbol{y}|^2\right) \exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)\right\} && (12) \end{aligned}$$

in $\mathbb{R}^n \times (0, T]$*. This can also be written as the logarithm of the expectation of the following Gaussian density*

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}} \cdot \log \mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)\right]. \tag{13}$$

Lemma 3.3 (Spatial gradients of the HJ solution)**.** *The spatial gradient to the HJ value function $\boldsymbol{v}(t;\boldsymbol{x})$ admits the following form*

$$D\boldsymbol{v}^\delta = \frac{1}{t\delta \boldsymbol{c}}\left(\boldsymbol{x} - \frac{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\boldsymbol{y}\cdot\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}\right) \tag{14}$$

where $\boldsymbol{y}$ is drawn from a Gaussian distribution $\mathcal{N}(\boldsymbol{x},\delta t)$, i.e. , with mean $\boldsymbol{x}$ and variance δt.

3.2. Relations to Backward Reachable Sets/Tubes

Since we now have the derivatives that constitute the viscous HJ equation, it is easy for us to rewrite (HJI-RCBRT) as

$$\begin{aligned} &\boldsymbol{v}_t^\delta(t;\boldsymbol{x}) + \min\{0, \boldsymbol{H}^\delta - \frac{\delta}{2}\Delta\boldsymbol{v}^\delta\} = 0, \\ &\boldsymbol{v}^\delta(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}). \end{aligned} \tag{HJI-RCBRT-Visc}$$

so that the solution to (HJI-RCBRT-Visc) is from integrating the quantities given in Lemma 3.3 is tantamount to sampling from the kernels of the given Gaussian densities. The invariant set $\mathcal{L}_0$ obtained at T is

$$\mathcal{L}_0(T) = \{\boldsymbol{x}\in\mathbb{R}^n \mid \boldsymbol{v}^\delta(0;\boldsymbol{x}) \le 0\}, \tag{15}$$

and the RCBRT is as given in (HJI-RCBRT). Similarly, the viscous version of the *reach-avoid BRT* i.e. (HJI-RCBRAT) can be represented as

$$\begin{aligned} &\min\{\boldsymbol{v}_t^\delta(t;\boldsymbol{x}) + \boldsymbol{H}^\delta - \frac{\delta}{2}\Delta\boldsymbol{v}^\delta, \boldsymbol{g}(t;\boldsymbol{x}) - \boldsymbol{\ell}(t;\boldsymbol{x})\} \le 0, \\ &\boldsymbol{v}^\delta(0;\boldsymbol{x}) = \boldsymbol{g}(0;\boldsymbol{x}). \end{aligned} \tag{HJI-RCBRAT-Visc}$$

3.3. Weighted Importance Sampling of Reachable Sets

We now have the components of the initial HJ problem (HJ-Visc) and the variational HJ PDEs (HJI-RCBRT-Visc), (HJI-RCBRAT-Visc) as Gaussian processes. While we may not recover the exact value function system, the mean and variance of its derivatives can be evaluated with the kernel expectations of Lemma 3.2, 3.3, and **??**. Previously, in high dimensions self-supervised deep learning has been employed for learning parameterized PDE value and value gradients approximations (Bansal & Tomlin, 2021). However, this requires matching the deep learning-based value function approximation to the levelset numerically computed value function in order to guide the solution. This inherently provides a limitation on the applications of such methods since levelset methods are inherently limiting in their scalability (Molu, 2024b; 2025); self-supervision with the original PDE's solution is only accurate up to a scale of the problem dimensions (reported to be 9D/10D in (Bansal & Tomlin, 2021)).

Algorithm 1 Quasi-Linearization Algorithm Cole-Hopf

Fix: $\epsilon > 0$, $\boldsymbol{v}^{(0)}(t;\boldsymbol{x}) = g(\boldsymbol{x})$, $\boldsymbol{c}^{(0)} = \frac{2\,\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{g})}{\delta\,|D\boldsymbol{g}|^2}$.
For $k = 0, 1, 2, \ldots$**:**

1. Freeze $\boldsymbol{c}^{(k)}$ at the current iterate.
2. Solve the heat equation $\boldsymbol{\omega}_t = \frac{\delta}{2}\Delta\boldsymbol{\omega}$ with initial data $\boldsymbol{\omega}^{(k)}(0;\boldsymbol{x}) = e^{-c^{(k)}g(\boldsymbol{x})}$.
3. Recover $\boldsymbol{v}^{(k+1)} = -(1/\boldsymbol{c}^{(k)})\log\boldsymbol{\omega}^{(k+1)}$.
4. Update $D\boldsymbol{v}^{(k+1)}$ and $\boldsymbol{c}^{(k+1)} = \frac{2\boldsymbol{H}(t;\boldsymbol{x},D\boldsymbol{v}^{(k+1)})}{\delta|D\boldsymbol{v}^{(k+1)}|^2}$.
5. Check convergence: $\|\boldsymbol{v}^{(k+1)} - \boldsymbol{v}^{(k)}\|/\|\boldsymbol{v}^{(k)}\| < \varepsilon$.

We propose a sequential importance sampling (IS) scheme for a high-dimensional $\boldsymbol{x}$ and consequently $\boldsymbol{v}^\delta$ can be devised for the target density $\mathbb{P}_{y\sim\mathcal{N}(\boldsymbol{x},\delta t)}$ to induce a chain-like decomposition of $\boldsymbol{v}^\delta := (\boldsymbol{v}_1^\delta, \cdots, \boldsymbol{v}_d^\delta), 1 \le d \le T$ i.e. ,

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = \boldsymbol{v}^\delta(d_1;\boldsymbol{x}_1)\prod_{d=2}^{T}\boldsymbol{v}^\delta(d;\boldsymbol{x}_d)|\boldsymbol{v}^\delta(d;\boldsymbol{x}_{[1:d-1]}) \tag{16}$$

from which we can build an importance sampling such as

$$q^\delta(t;\boldsymbol{x}) = q_1^\delta(d_1;\boldsymbol{x}_1)\prod_{d=2}^{T}q_d^\delta(d;\boldsymbol{x}_d)|q_d^\delta(d;\boldsymbol{x}_{[1:d-1]}). \tag{17}$$

In this sentiment, we give the following corollaries to Lemmas 3.2 and 3.3.

Corollary 3.4. *For the* (HJI-RCBRT-Visc) *and* (HJI-RCBRAT-Visc), $\boldsymbol{g}(\boldsymbol{y})$ *the above equation transforms to $\boldsymbol{g}(\boldsymbol{x} + \sqrt{\delta(T-t)}\boldsymbol{y}_i)$. For numerical stability, we use the log-sum-exp identity*

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = \frac{-1}{\boldsymbol{c}}\log\frac{1}{N}\sum_{i=1}^{N}\exp(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{x} + \sqrt{\delta(T-t)}\boldsymbol{y}_i)), \tag{18}$$

with $\boldsymbol{y}_i \sim \mathcal{N}(\boldsymbol{x},\delta t)$

Corollary 3.5. *As a Monte Carlo approximation with samples $\boldsymbol{d}_1,\ldots,\boldsymbol{d}_N$ where $\boldsymbol{d}_i = \boldsymbol{x} + \sqrt{\delta(T-t)}\boldsymbol{y}_i$, with $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{x},\delta t)$ the importance-weighted factorization of* (14) *can be expressed as*

$$D\boldsymbol{v}^\delta(t;\boldsymbol{x}) = \frac{1}{t\delta\boldsymbol{c}}\left(\boldsymbol{x} - \frac{\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{d}_i\cdot\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{d}_i))}{\frac{1}{N}\sum_{i=1}^{N}\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{d}_i))}\right). \tag{19}$$

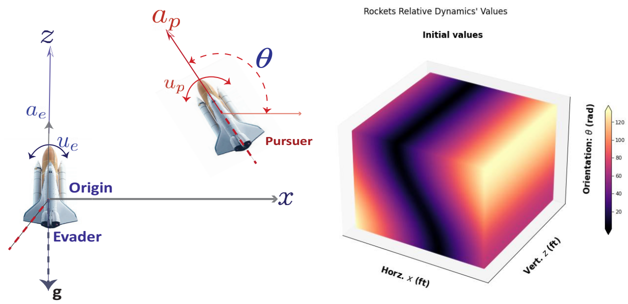

Figure 1. Left: Motion of two rockets on a Cartesian $\boldsymbol{xz}$-plane with a thrust inclination in relative coordinates given by $\theta := u_p - u_e$. *Right:* Representation of the initial values as a heat map for the two rockets described in (25).

We state our quasi-linearization iteration scheme in Algorithm 1 for recovering the backward reachable sets/tubes based on the Gaussian expectation formulas computed in Lemmas 3.2, 3.3. The heat equation solution via Gaussian expectations is estimated via unbiased Monte Carlo. In the next section, we introduce numerical examples to test the veracity of our proposals.

4. Numerical Results

In this section, we provide a representative problem and amend it to the viscous HJI PDE (HJI-RCBRT-Visc) to test and verify our hypothesis.

4.1. The Rockets Launch Problem

We consider the rocket launch problem of Dreyfus (Dreyfus, 1966) which was amended to a differential game between two identical rockets, $\boldsymbol{P}$ and $\boldsymbol{E}$, on an (x, z) cross-section of a Cartesian plane in (Molu, 2024a; 2025). We want to compute the usable part (Isaacs, 1999) or RCBRT (Mitchell et al., 2005) of the *approximate* terminal surface's boundary for a predefined target set over a target tube. The usable part entails the regions of the state-space for which the min-max operation over either strategy of $\boldsymbol{P}$ and $\boldsymbol{E}$ is below zero and where the boundary of the usable part (BUP) constitute where (HJI-RCBRAT) is exactly zero.

For a two-player differential game, let $\boldsymbol{P}$ and $\boldsymbol{E}$ share identical dynamics in a general sense so that we can freely choose the coordinates of $\boldsymbol{P}$; however, $\boldsymbol{E}$'s origin is a distance Φ away from (x, z) at plane's origin (see Fig. 1) so that the $\boldsymbol{PE}$ vector's inclination measured counterclockwise from the $\boldsymbol{x}$ axis is θ.

Hence, we may consider the current problem as a terminal pursuit-evasion *differential* game, whereupon the game terminates when *capture* occurs. This eventual terminal represents the final *overapproximated* (non)-capture surface for the $\boldsymbol{P} - \boldsymbol{E}$ game (to be shortly introduced). We desire that every `max` and `min` operations during a *game of kind* are in the interior of the plane where motion exists; this way, no sudden changes from extremes are too aggravating in cost.

Let the states of $\boldsymbol{P}$ and $\boldsymbol{E}$ be $(\boldsymbol{x}_p, \boldsymbol{x}_e)$, respectively, driven by thrusts (u_p, u_e) respectively (see Figure 1). Fix the rockets' range so that what is left of the motion of either $\boldsymbol{P}$ or $\boldsymbol{E}$'s is restricted to orientation on the (x, z) plane as illustrated in Fig. 1. where a and g are respectively the acceleration and gravitational accelerations (in feet per square second) [2]. As long as $\boldsymbol{E}$ remains within this target region or *backward reachable tube* (or BRT), $\boldsymbol{P}$ cannot cause damage or exercise an action of deleterious consequence on, say, the territory being guarded by $\boldsymbol{E}$. Setting up $\boldsymbol{E}$ to maximize a payoff quantity with the largest possible margin or at least frustrate the efforts of $\boldsymbol{P}$ with minimal collateral damage while the pursuer minimizes this quantity constitutes a terminal value *optimal* differential game: there is no optimal pursuit without an optimal evasion.

The motion of $\boldsymbol{P}$ relative to $\boldsymbol{E}$'s along the $(\boldsymbol{x}, \boldsymbol{z})$ plane includes the relative orientation, the control input, shown in Fig. 1 as $\theta = u_p - u_e$. Following the conventions in Fig. 1, the game's relative equations of motion in *reduced space* (Isaacs, 1999, §2.2) was derived as (see (Molu, 2024a, §VI.A)) is $\boldsymbol{x} = (x, z, \theta)$ where $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $(x, z) \in \mathbb{R}^2$ are

$$\dot{\boldsymbol{x}} := \begin{cases} \dot{x} & = a_p \cos\theta + u_e x, \\ \dot{z} & = a_p \sin\theta + a_e + u_e x - g, \\ \dot{\theta} & = u_p - u_e. \end{cases} \tag{20}$$

The capture radius of the origin-centered circle Φ (we set $\boldsymbol{\ell} = 1.5$ ft) is $\|\boldsymbol{PE}\|_2$ so that $\boldsymbol{\ell}^2 = x^2 + z^2$. All capture points are specified by the variational HJ PDE (Mitchell et al., 2005):

$$\boldsymbol{v}_t^\delta(t; \boldsymbol{x}) + \min\left[0, \boldsymbol{H}(t; \boldsymbol{x}, D\boldsymbol{\ell}) - \frac{\delta}{2}\Delta \boldsymbol{v}^\delta\right] = 0, \tag{21}$$

with Hamiltonian given by

$$\boldsymbol{H}(t; \boldsymbol{x}, p) = -\max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p].} \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix} \begin{bmatrix} a_p \cos\theta + u_e x \\ a_p \sin\theta + a_e + u_p x - g \\ u_p - u_e \end{bmatrix}. \tag{22}$$

Here, p are the co-states, and $[\underline{u}_e, \bar{u}_e]$ denotes extremals that the evader must choose as input in response to the extremal controls that the pursuer plays i.e. $[\underline{u}_p, \bar{u}_p]$. Rather than resort to analytical *geometric reasoning*, we may analyze possibilities of behavior by either agent via a principled

[2] We set $a = 1 ft/sec^2$ and $g = 32 ft/sec^2$ in our simulation.

numerical simulation. This is the essence of this work. From (22), set $\underline{u}_e = \underline{u}_p = \underline{u} \triangleq -1$ and $\bar{u}_p = \bar{u}_e = \bar{u} \triangleq +1$ so that $\boldsymbol{H}(t; \boldsymbol{x}, p)$ is

$$
\begin{aligned}
&- \max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} \begin{bmatrix} p_1(a_p \cos\theta + u_e x) + \\ p_2(a_p \sin\theta + a_e + \\ u_p x - g) + p_3(u_p - u_e) \end{bmatrix} \\
&\triangleq -a p_1 \cos\theta - p_2(g - a - a\sin\theta) - \bar{u}|p_1 x + p_3| \\
&\qquad + \underline{u}|p_2 x + p_3|, \qquad (23)
\end{aligned}
$$

where the last line follows from setting $a_e = a_p \triangleq a$. Thus, the rockets HJI equation (21) becomes

$$
\begin{aligned}
&\boldsymbol{v}_t^\delta + \min\left[0, (-a p_1 \cos\theta - p_2(g - a - a\sin\theta) - \right. \\
&\qquad \left. |p_1 x + p_3| - |p_2 x + p_3|) - \frac{\delta}{2}\Delta \boldsymbol{v}^\delta\right] = 0, \quad (24)
\end{aligned}
$$

where p_1, p_2, p_3 are the coefficients of the co-states. A similar derivation can be obtained for the (HJI-RCBRAT). In our experiments, we set $a = 1 ft/sec^2$ and $g = 32 ft/sec^2$.

4.2. Discussion

We take the infinitesimal physics problem above and make it algorithmic first by discretizing the spatial and temporal domains i.e.

$$
(x, z) \in (-100, +100],\ \theta \in (-\pi/2, \pi/2],\ t \in (0, 1], \qquad (25)
$$

with 1000 points along the temporal dimension and $\boldsymbol{x}_k - \boldsymbol{x}_{k-1} = 1/100$ along the spatial dimensions, where $k \in \mathbb{N}$. All our codes are implemented in pytorch, that runs on an Azure Linux node with the cpu processor specification: Intel(R) Core(TM) i7-14700K at 5.6GHz and 31GiB memory; The pytorch runs on a Ubuntu 22.04 operating system.

We thus demonstrate that the (i) introduced Cole-Hopf transformation (Linear-Trans) is valid for viscous HJ PDEs; (ii) frozen approximation coefficient (9) converges in the limit of the iteration; (iii) heat equation solution via the Gaussian kernel transformation (13) is exact; (iv) Monte Carlo sampling provides unbiased estimates; and (v) quasi-linearization fixed-point iteration is well-posed.

We defined the target set as the ${}_2$ distance $\ell(\boldsymbol{x}) = \|x^2 + z^2\|_2$ and then run the Algorithm 1 on a time range of $(0, 1]$ seconds; we chose Dirichlet boundary conditions

$$
\boldsymbol{v}^\delta(0; \boldsymbol{x}) = \boldsymbol{g}(0; \boldsymbol{x}) \triangleq 0. \qquad (26)
$$

The spatial gradients (co-states) were computed using the transformation (13) and (14) and the Hamiltonian was iteratively solved for 8,000 samples according to Alg 1.

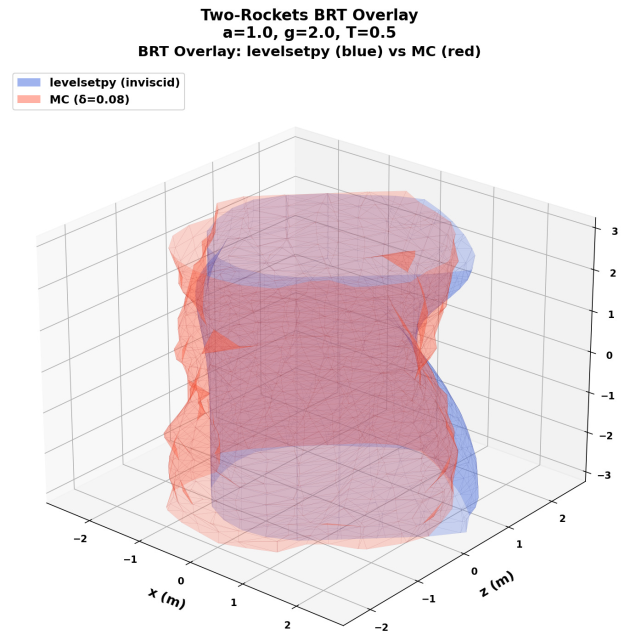

Figure 2. Rockets BRT comparison: Levelsetpy generated solution cyan and our Monte Carlo Sampling Scheme pink.

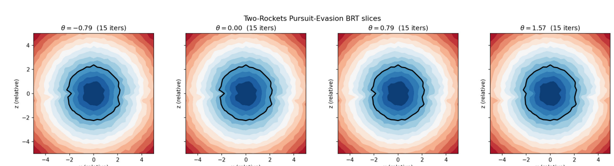

Figure 3. Rockets BRT Slices

5. Conclusion

We have presented a Gaussian-approximation of the HJI PDE employed in computing the safe reachable sets of dynamical systems. Leveraging a linear transformation of the variational HJI PDE of (Mitchell et al., 2005), we expressed the value function, its gradients and Hamiltonian as expectation of Gaussian random variables of the state and time. This allows the recovery of the exact value function system, the mean and variance of its derivatives — which can then be evaluated with the kernel expectations of Lemma 3.2, 3.3, and **??**. Interesting questions pop up with this new formulation. Scalability of HJ reachable sets is within reach with appropriate approximations of the kernels that constitute the viscous HJI PDE.

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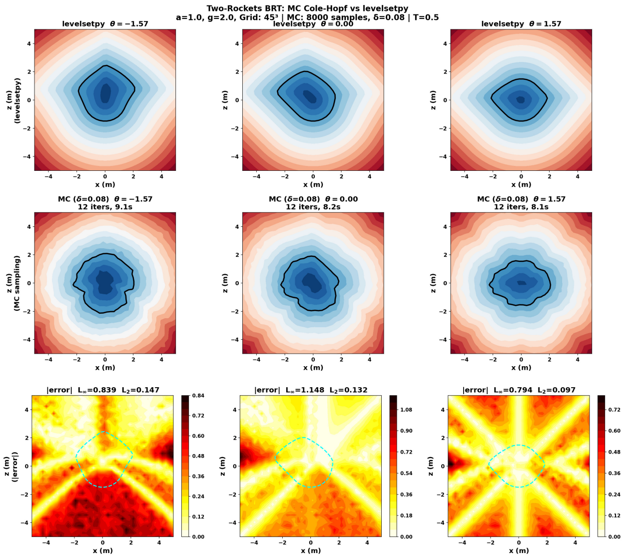

Figure 4. Rockets BRT Slices: Levelsetpy vs Monte-Carlo (Ours). The bottom row shows a heat map of the errors at various relative vehicle orientations (the control input) between the Monte Carlo sampling scheme and the levelsetpy scheme.

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A. Background and Preliminaries (Unabridged).

This section provides a context for the contribution of this work.

A.1. Dynamic Programming and Two-Person Games.

The formal relationships between the dynamic programming (DP) optimality condition for the *value* in differential two-person zero-sum games, and the solutions to PDEs that solve "min-max" or "max-min" type nonlinearity (the Isaacs' equation) was presented in (Isaacs, 1999). Essentially, Isaacs' claim was that if the *value* functions are smooth enough, then they solve certain first-order partial differential equations (PDE) problems with "max-min" or "min-max"-type nonlinearity. However, the DP value functions are seldom regular enough to admit a solution in the classical sense. "Weaker" solutions, on the other hand (Lions, 1982; Evans & Souganidis, 1984a; Crandall et al., 1984; Crandall & Majda, 1980b; Evans & Souganidis, 1984b), provide generalized "viscosity" solutions to HJ PDEs under relaxed regularity conditions; these viscosity solutions are not necessarily differentiable anywhere in the state space, and the only regularity prerequisite in the definition is continuity (Crandall & Lions, 1983). However, wherever they are differentiable, they satisfy the upper and lower values of HJ PDEs (discussed in § A.2) in a classical sense. Thus, they lend themselves well to many real-world problems existing at the interface of discrete, continuous, and hybrid systems (Lygeros, 2004; Osher & Sethian, 1988; Mitchell, 2020; Evans & Souganidis, 1984b; Mitchell et al., 2005).

Matter-of-factly, viscosity solutions to *Cauchy-type*[3] HJ Equations are highly useful in backward reachability analysis (Mitchell et al., 2005). For a state $\boldsymbol{x} \in \Omega$ and a fixed time t: $0 \le t < T$, suppose that the set of all controls for players $\boldsymbol{P}$ and $\boldsymbol{E}$ are respectively

$$\bar{\mathcal{U}} \equiv \{\boldsymbol{u} : [t, T] \to \mathcal{U} | \boldsymbol{u} \text{ measurable}, \mathcal{U} \in \mathbb{R}^m\}, \quad \bar{\mathcal{W}} \equiv \{\boldsymbol{w} : [t, T] \to \mathcal{V} | \boldsymbol{v} \text{ measurable}, \mathcal{W} \subset \mathbb{R}^p\}. \tag{A.1}$$

We are concerned with the differential equation,

$$\dot{\boldsymbol{x}}(\tau) = f(\tau, \boldsymbol{x}(\tau), \boldsymbol{u}(\tau), \boldsymbol{w}(\tau)), \;\; \boldsymbol{x}(t) = \boldsymbol{x}, \;\; T \le \tau \le t \tag{A.2}$$

where $f(\tau, \cdot, \cdot, \cdot)$ and $\boldsymbol{x}(\cdot)$ are bounded and Lipschitz continuous. This bounded Lipschitz continuity property assures uniqueness of the system response $\boldsymbol{x}(\cdot)$ to controls $\boldsymbol{u}(\cdot)$ and $\boldsymbol{v}(\cdot)$ (Evans & Souganidis, 1984b). Associated with (A.2) is the payoff functional

$$\boldsymbol{\ell}(\boldsymbol{u}, \boldsymbol{w}) = \boldsymbol{\ell}(t; \boldsymbol{x}, \boldsymbol{u}(\cdot), \boldsymbol{w}(\cdot)) \; \equiv \int_t^T l(\tau, \boldsymbol{x}(\tau), \boldsymbol{u}(\tau), \boldsymbol{w}(\tau)) d\tau + \boldsymbol{g}(\boldsymbol{x}(T)), \tag{A.3}$$

where $\boldsymbol{g}(\cdot) : \mathbb{R}^n \to \mathbb{R}$ satisfies

$$|\boldsymbol{g}(\boldsymbol{x})| \le k_1, \quad |\boldsymbol{g}(\boldsymbol{x}) - \boldsymbol{g}(\hat{\boldsymbol{x}})\, | \le k_1 |\boldsymbol{x} - \hat{\boldsymbol{x}}\, | \tag{A.4}$$

and $l : [0, T] \times \mathbb{R}^n \times \mathcal{U} \times \mathcal{W} \to \mathbb{R}$ is bounded and uniformly continuous, with

$$|\; l(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) \;| \le k_2, \; |\; l(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}) - l(t; \hat{\boldsymbol{x}}, \boldsymbol{u}, \boldsymbol{w}) \;| \le k_2 \;|\; \boldsymbol{x} - \hat{\boldsymbol{x}} \;| \tag{A.5}$$

for constants k_1, k_2 and all $0 \le t \le T$, $\hat{\boldsymbol{x}}$, $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{u} \in \mathcal{U}$ and $\boldsymbol{w} \in \mathcal{W}$. We call T the *terminal time* (it may be infinity!) and the integral, when it does not depend on the control laws, is the *performance index*. The evader's goal is to maximize the payoff (A.3) and pursuer's goal is to minimize it.

A.2. Upper and Lower Values of the Differential Game.

Suppose that the pursuer's mapping strategy (starting at t) is $\beta : \bar{\mathcal{U}}(t) \to \bar{\mathcal{W}}(t)$ provided for each $t \le \tau \le T$ and $\boldsymbol{u}, \hat{\boldsymbol{u}} \in \bar{\mathcal{U}}(t)$; then $\boldsymbol{u}(\bar{t}) = \hat{\boldsymbol{u}}(\bar{t})$ a.e. on $t \le \bar{t} \le \tau$ implies $\beta[\boldsymbol{u}](\bar{t}) = \beta[\hat{\boldsymbol{u}}](\bar{t})$ a.e. on $t \le \bar{t} \le \tau$. The differential game's lower value for a solution $\boldsymbol{x}(t)$ that solves (A.2) for $\boldsymbol{u}(t)$ and $\boldsymbol{v}(t) = \beta[\boldsymbol{u}](\cdot)$ is

$$\boldsymbol{v}^-(t; \boldsymbol{x}) = \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \boldsymbol{\ell}(\boldsymbol{u}, \beta[\boldsymbol{u}]) \triangleq \inf_{\beta \in \mathcal{B}(t)} \sup_{\boldsymbol{u} \in \mathcal{U}(t)} \int_t^T l(\tau, \boldsymbol{x}(\tau), \boldsymbol{u}(\tau), \beta[\boldsymbol{u}](\tau)) d\tau + \boldsymbol{g}\left(\boldsymbol{x}(T)\right). \tag{A.6}$$

[3] Cauchy-type HJ equations are time-dependent versions of the HJ PDE.

Similarly, suppose that the evader's mapping strategy (starting at t) is $\alpha : \bar{\mathcal{W}}(t) \to \bar{\mathcal{U}}(t)$ provided for each $t \le \tau \le T$ and $\boldsymbol{w}, \hat{\boldsymbol{w}} \in \bar{\mathcal{W}}(t)$; then $\boldsymbol{w}(\bar{t}) = \hat{\boldsymbol{w}}(\bar{t})$ a.e. on $t \le \bar{t} \le \tau$ implies $\alpha[\boldsymbol{w}](\bar{t}) = \alpha[\hat{\boldsymbol{w}}](\bar{t})$ a.e. on $t \le \bar{t} \le \tau$. The differential game's upper value for a solution $\boldsymbol{x}(t)$ that solves (A.2) for $\boldsymbol{u}(t) = \alpha[\boldsymbol{w}](\cdot)$ and $\boldsymbol{w}(t)$ is

$$\boldsymbol{v}^+(t;\boldsymbol{x}) = \sup_{\alpha \in \mathcal{A}(t)} \inf_{\boldsymbol{w} \in \mathcal{W}(t)} \boldsymbol{\ell}(\alpha[\boldsymbol{w}], \boldsymbol{w}) \triangleq \sup_{\alpha \in \mathcal{A}(t)} \inf_{w \in \mathcal{W}(t)} \int_t^T l(\tau, \boldsymbol{x}(\tau), \alpha[\boldsymbol{w}](\tau), \boldsymbol{w}(\tau)) d\tau + \boldsymbol{g}\left(\boldsymbol{x}(T)\right). \tag{A.7}$$

These non-local PDEs ((A.6) and (A.7)) are hardly smooth throughout the state space so that they lack classical solutions even for smooth Hamiltonian and boundary conditions. However, these two values are "viscosity" (generalized) solutions (Lions, 1982; Crandall & Lions, 1983) of the associated HJ-Isaacs (HJI) PDE, i.e. solutions which are *locally Lipschitz* in $\Omega \times [0, T]$, and with at most first-order partial derivatives in the Hamiltonian.

A.3. Viscosity Solution of HJ-Isaac's Equations.

For any optimal control problem a value function is constructed based on the optimal cost (or payoff) of any input phase $(\boldsymbol{x}, T)$. In reachability analysis, typically this is defined using a terminal cost function $g(\cdot) : \mathbb{R}^n \to \mathbb{R}$ that satisfies

$$|\boldsymbol{g}(\boldsymbol{x})| \le k, \;\; |\boldsymbol{g}(\boldsymbol{x}) - \boldsymbol{g}(\hat{\boldsymbol{x}})\; | \le k|\boldsymbol{x} - \hat{\boldsymbol{x}}\;| \tag{A.8}$$

for constant k and all $T \le t \le 0$, $\hat{\boldsymbol{x}}$, $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{u} \in \mathcal{U}$ and $\boldsymbol{w} \in \mathcal{W}$. The zero sublevel set of $\boldsymbol{g}(\boldsymbol{x})$ i.e.

$$\mathcal{L}_0 = \{\boldsymbol{x} \in \bar{\Omega} \,|\, \boldsymbol{g}(\boldsymbol{x}) \le 0\}, \tag{A.9}$$

Lemma A.1. *The lower value $\boldsymbol{v}^-$ in (4a) is the viscosity solution to the lower Isaac's equation*

$$\boldsymbol{v}_t^- + \boldsymbol{H}^-(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},D\boldsymbol{v}^-) = 0, \;\; t \in [0,T] \;\; \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{v}^-(T;\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}(T)), \quad \boldsymbol{x} \in \mathbb{R}^m \tag{A.10}$$

with lower Hamiltonian,

$$\boldsymbol{H}^-(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p) = \max_{u \in \mathcal{U}} \min_{w \in \mathcal{W}} \langle \boldsymbol{f}(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}), p \rangle. \tag{A.11}$$

where p, the co-state, is the spatial derivative of $\boldsymbol{v}^-$ w.r.t $\boldsymbol{x}$.

Lemma A.2. *The upper value $\boldsymbol{v}^+$ in (A.7) is the viscosity solution of the upper Isaac's equation*

$$\boldsymbol{v}_t^+ + \boldsymbol{H}^+(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},D\boldsymbol{v}^+) = 0, \; t \in [0,T]\,, \; \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{v}^+(T;\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}(T)), \quad \boldsymbol{x} \in \mathbb{R}^n \tag{A.12a}$$

with upper Hamiltonian,

$$\boldsymbol{H}^+(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p) = \min_{\boldsymbol{w} \in \mathcal{W}} \max_{\boldsymbol{u} \in \mathcal{U}} \langle \boldsymbol{f}(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w}), p \rangle, \tag{A.13}$$

with p being appropriately defined.

Corollary A.3. *(i) $\boldsymbol{v}^- \le \boldsymbol{v}^+$ over $(t \in [0,T]\; \boldsymbol{x} \in \mathbb{R}^n)$ (ii) if for all $t \in [0,T]\,, (\boldsymbol{x}, p) \in \mathbb{R}^n$, the minimax condition is satisfied i.e. $\boldsymbol{H}^+(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p) = \boldsymbol{H}^-(t;\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},p)$, then $\boldsymbol{v}^- \equiv \boldsymbol{v}^+$.*

A.4. Reachability for Systems Verification.

Reachability analysis is one of many verification methods that allows us to reason about (control-affine) dynamical systems. The verification problem may consist in finding a *set of reachable states* that lie along the trajectory of the solution to a first order nonlinear partial differential equation that originates from some initial state $\boldsymbol{x}_0 = \boldsymbol{x}(0)$ up to a specified time bound, $t = t_f$. *From a set of initial and unsafe state sets, the time-bounded safety verification problem is to determine if there is an initial state and a particular time within the bound that the solution to HJI PDE enters the unsafe set*.

Reachability could be analyzed in a (i) *forward* sense, whereupon system trajectories are examined to determine if they enter certain states from an *initial set*; (ii) *backward* sense, whereupon system trajectories are examined to determine if they enter certain *target sets*; (iii) *reach set* sense, in which they are examined to see if states reach a set at a *particular time*; or (iv) *reach tube* sense, in which they are evaluated that they reach a set at a point *during a time interval*.

Backward reachability consists in avoiding an unsafe set of states under the worst-possible disturbance at all times; relying on nonanticipative control strategies. Backward reachable sets (BRS) and backward reachable tubes (BRTs) are popularly analyzed in a game of two vehicles with non-stochastic dynamics (Merz, 1972). Such BRTs possess discontinuity at cross-over points (which exist at edges) on the surface of the tube, and may be non-convex. Therefore, treating the end-point constraints under these discontinuity characterizations need careful consideration and analysis when switching control laws if the underlying PDE does not have continuous partial derivatives (we discuss this further in § 3).

A.5. Reachability from Differential Games Optimal Control

For any admissible control-disturbance pair $(\boldsymbol{u}(\cdot), \boldsymbol{w}(\cdot))$ and initial phase $(\boldsymbol{x}_0, t_0)$, Crandall (Crandall & Lions, 1983) and Evan's (Evans & Souganidis, 1984a) claim is that there exists a unique function

$$\boldsymbol{\xi}(t) = \boldsymbol{\xi}(t; t_0, \boldsymbol{x}_0, \boldsymbol{u}(\cdot), \boldsymbol{w}(\cdot)) \tag{A.14}$$

that satisfies (A.2) a.e. with the property that

$$\boldsymbol{\xi}(t_0) = \boldsymbol{\xi}(t_0; t_0, \boldsymbol{x}_0, \boldsymbol{u}(\cdot), \boldsymbol{w}(\cdot)) = \boldsymbol{x}_0. \tag{A.15}$$

Read (A.14): the motion of (A.2) passing through phase $(\boldsymbol{x}_0, t_0)$ under the action of control $\boldsymbol{u}$, and disturbance $\boldsymbol{w}$, and observed at a time t afterwards. One way to design a system verification problem is compute the reachable set of states that lie along the trajectory (A.14) such that we evade the unsafe sets up to a time e.g. t_f within a given time bound e.g. $[t_0, t_f]$. In this regard, we discard the *cost-to-go*, $l(t; \boldsymbol{x}(\tau), \boldsymbol{u}(\tau), \boldsymbol{w}(\tau))$ in (A.3), (4a), or (A.7) and certify safety as resolving the terminal value, $g(\boldsymbol{x}(T))$.

In backward reachability analysis, the lower value of the differential game is used in constructing an analysis of the backward reachable set (or tube). Therefore, we can cast a target set as the time-resolved terminal value $\boldsymbol{v}^-(\boldsymbol{x}, T) = \boldsymbol{g}(\boldsymbol{x}(T))$ so that given a time bound, and an unsafe set of states, the time-bounded safety verification problem consists in certifying that there is no phase within the target set (Target-Set) such that the solution to (A.2) enters the unsafe set. Following the backward reachability formulation of (Mitchell et al., 2005), we say the zero sublevel set of $g(\cdot)$ in (A.10) i.e.

$$\mathcal{L}_0 = \{\boldsymbol{x} \in \bar{\Omega} \,|\, \boldsymbol{g}(\boldsymbol{x}) \leq 0\}, \tag{A.16}$$

is the *target set* in the phase space $\Omega \times \mathbb{R}$ for a backward reachability problem (proof in (Mitchell et al., 2005)). This target set can represent the failure set, regions of danger, or obstacles to be avoided etc in the state space. Note that the target set, $\mathcal{L}_0$, is a closed subset of $\mathbb{R}^n$ and is in the closure of Ω. And the *robustly controlled backward reachable tube* for $\tau \in [-T, 0]$[4] is the closure of the open set

$$\mathcal{L}([\tau, 0], \mathcal{L}_0) = \{\boldsymbol{x} \in \Omega \,|\, \exists\, \beta \in \bar{\mathcal{W}}(t)\, \forall\, \boldsymbol{u} \in \mathcal{U}(t), \exists\, \bar{t} \in [-T, 0], \boldsymbol{\xi}(\bar{t}) \in \mathcal{L}_0\},\ \bar{t} \in [-T, 0]\,. \tag{A.17}$$

Read: the set of states from which the strategies of $\boldsymbol{P}$, and for all controls of $\boldsymbol{E}$ imply that we reach the target set within the interval $[T, 0]$. More specifically, following Lemma 2 of (Mitchell et al., 2005), the states in the reachable set admit the following properties w.r.t the value function $\boldsymbol{v}$

$$\boldsymbol{x} \in \mathcal{L}_0 \implies \boldsymbol{v}^-(\boldsymbol{x}, t) \leq 0, \quad \boldsymbol{v}^-(\boldsymbol{x}, t) \leq 0 \implies \boldsymbol{x} \in \mathcal{L}_0. \tag{A.18a}$$

Observe: (a) The goal of the pursuer, or $\boldsymbol{P}$, is to drive the system's trajectories into the unsafe set i.e., $\boldsymbol{P}$ has $\boldsymbol{u}$ at will and aims to minimize the termination time of the game (c.f. (Target-Set)); (b) the evader, or $\boldsymbol{E}$, seeks to avoid the unsafe set i.e., $\boldsymbol{E}$ has controls $\boldsymbol{w}$ at will and seeks to maximize the termination time of the game (c.f. (Target-Set)); (c) $\boldsymbol{E}$ has regular controls, $\boldsymbol{u}$, drawn from a Lebesgue measurable set, $\mathcal{U}$ (c.f. (4a)). (d) $\boldsymbol{P}$ possesses *nonanticipative strategies* (c.f. (4a)) i.e. $\beta[\boldsymbol{u}](\cdot)$ such that for any of the ordinary controls, $\boldsymbol{u}(\cdot) \in \mathcal{U}$ of $\boldsymbol{E}$, $\boldsymbol{P}$ knows how to optimally respond to $\boldsymbol{E}$'s inputs. This is a classic reachability problem on the resolution of the infimum-supremum over the *strategies* of $\boldsymbol{P}$ and *controls* of $\boldsymbol{E}$ with the time of capture resolved as an extremum of a cost functional) over a time interval.

The terminal value $\boldsymbol{v}^-(t; \boldsymbol{x})$, that characterizes the target set $\mathcal{L}_0$ is the viscosity solution to the HJI PDE

$$\boldsymbol{v}_t^-(t; \boldsymbol{x}) + \min\{0, \boldsymbol{H}^-(t; \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{w}, D\boldsymbol{v}^-)\} = 0, \quad \boldsymbol{v}^-(0; \boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}), \tag{A.19}$$

where the vector field $\boldsymbol{v}_{\boldsymbol{x}}^-$ is known in terms of the game's terminal conditions so that the overall game is akin to a two-point boundary-value problem. For ease of readability, we will remove the negative superscript on the lower value and Hamiltonian (5).

[4] The (backward) horizon, $-T$ is negative for $T > 0$.

B. HJ PDE Linearization

In this section, we construct the Cole-Hopf-like linearization of the viscous Hamilton-Jacobi (HJ) PDE and propose a sampling machinery we use in building our results. We show that the transformation is *exact* only when the Hamiltonian is quadratic in the co-state i.e. $\boldsymbol{H} = \frac{1}{2}\langle \boldsymbol{p}^\top, \boldsymbol{p}\rangle$, and we derive the precise residual that for general Hamiltonians. We then formulate the quasi-linearization iterative algorithm for computing (HJI-RCBRT-Visc) and (HJI-RCBRAT-Visc). We finish this appendix with the correct Gaussian expectation formulas for the value function and its spatial gradient.

B.1. The viscous HJ Equation's solution

In these proofs that follow, we will express the solutions to the viscous HJ equation as the logarithm of the expectation of Gaussian kernels that parameterize the state space. Let us construct the spatial and time derivatives to the solution to the viscous intial-value-HJ problem (HJ-Visc).

Proof of Proposition 3.1. Recall from (HJ-Visc) that

$$\boldsymbol{v}_t^\delta + \boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta) = \frac{\delta}{2}\Delta \boldsymbol{v}^\delta \quad \text{in} \quad \mathbb{R}^n \times (0,T], \quad \boldsymbol{v}^\delta(0;\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{x}) \text{ on } \mathbb{R}^n \times \{t=0\}, \tag{B.1}$$

where $\delta > 0$ is the viscosity parameter, $\boldsymbol{H} : \mathbb{R}\times\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is the Hamiltonian, and $g : \mathbb{R}^n \to \mathbb{R}$ is the terminal/initial cost. Let $\boldsymbol{\omega}^\delta = \boldsymbol{\varphi}(\boldsymbol{v}^\delta(t;\boldsymbol{x}))$, where $\boldsymbol{\varphi} : \mathbb{R}^n \to \mathbb{R}$ is a smooth function. Our goal is to choose $\boldsymbol{\varphi}$ so that $\boldsymbol{\omega}^\delta$ solves a linear partial differential equation. Differentiating, we find that

$$\boldsymbol{\omega}_t^\delta = \boldsymbol{\varphi}'(\boldsymbol{v}^\delta)\boldsymbol{v}_t^\delta \text{ and } \Delta\boldsymbol{\omega}^\delta = \boldsymbol{\varphi}''(\boldsymbol{v}^\delta) \mid D\boldsymbol{v}^\delta \mid^2 + \boldsymbol{\varphi}'(\boldsymbol{v}^\delta)\Delta\boldsymbol{v}^\delta. \tag{B.2}$$

Thus,

$$\boldsymbol{\omega}_t^\delta = \boldsymbol{\varphi}'\boldsymbol{v}_t^\delta \triangleq \boldsymbol{\varphi}'(\boldsymbol{v}^\delta)\left(\frac{\delta}{2}\Delta\boldsymbol{v}^\delta - \boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta)\right), \tag{B.3a}$$

$$= \frac{\delta}{2}\left[\Delta\boldsymbol{\omega}^\delta - \boldsymbol{\varphi}''(\boldsymbol{v}^\delta) \mid D\boldsymbol{v}^\delta \mid^2\right] - \boldsymbol{\varphi}'(\boldsymbol{v}^\delta)\boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta), \tag{B.3b}$$

$$= \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta - \left(\frac{\delta}{2}\boldsymbol{\varphi}''(\boldsymbol{v}^\delta) \mid D\boldsymbol{v}^\delta \mid^2 + \boldsymbol{\varphi}'(\boldsymbol{v}^\delta)\boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta)\right). \tag{B.3c}$$

Suppose that we choose,

$$\boldsymbol{\varphi}^\delta = \exp\left(\frac{-2}{\delta}\cdot\frac{\boldsymbol{H}^\delta}{|D\boldsymbol{z}|^2}\boldsymbol{z}\right), \tag{B.4}$$

as the solution to the residual $\boldsymbol{R}(t;\boldsymbol{x}) := \frac{\delta}{2}\boldsymbol{\varphi}''(\boldsymbol{v}^\delta) \mid D\boldsymbol{v}^\delta \mid^2 + \boldsymbol{\varphi}'(\boldsymbol{v}^\delta)\boldsymbol{H}^\delta = 0$, it follows that

$$\boldsymbol{\omega}_t^\delta \triangleq \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta. \tag{B.5}$$

Set

$$\boldsymbol{c}(t;\boldsymbol{x}) = \tfrac{2}{\delta}\cdot\boldsymbol{H}^\delta/|D\boldsymbol{v}^\delta|^2 \tag{B.6}$$

and let $\boldsymbol{c}(t;\boldsymbol{x})$ be locally frozen as a spatially-varying coefficient rather than a functional of $\boldsymbol{v}^\delta$.

Corollary B.1. *Note that $\boldsymbol{R}(t;\boldsymbol{x})$ becomes zero when $\boldsymbol{\varphi}^\delta$ is a constant. This holds when $\boldsymbol{H}^\delta/|D\boldsymbol{v}^\delta|^2$ is independent of the phase $(\boldsymbol{x}, t)$. In these situations, $\boldsymbol{H}^\delta = \frac{1}{2}\langle\boldsymbol{p}^\top, \boldsymbol{p}\rangle$, $\boldsymbol{c} = 1/\delta = const.$ thus recovering Proposition 3.1.*

Then, dropping the templated arguments for ease of readability, if $\boldsymbol{v}^\delta$ solves (HJ-Visc), then we have the following approximate local transformation

$$\boldsymbol{\omega}^\delta = \exp\left(-\boldsymbol{c}\boldsymbol{v}^\delta\right) \tag{B.7}$$

solves the initial value problem for the differential equation

$$\boldsymbol{\omega}_t^\delta - \frac{\delta}{2}\Delta\boldsymbol{\omega}^\delta = \boldsymbol{R}(t;\boldsymbol{x}) \equiv 0 \quad \text{in} \quad \Omega\times(0,T], \text{ and } \boldsymbol{\omega}^\delta(0;\boldsymbol{x}) = \exp\left(-\boldsymbol{g}(0;\boldsymbol{x})\right) \quad \text{on} \quad \partial\Omega\times\{t=0\}, \tag{B.8}$$

where the **residual** is

$$\boldsymbol{R}(t;\boldsymbol{x}) = \left(\frac{\delta}{2}\boldsymbol{\varphi}''(\boldsymbol{v}^\delta) \mid D\boldsymbol{v}^\delta \mid^2 + \boldsymbol{\varphi}'(\boldsymbol{v}^\delta)\boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta)\right) \equiv 0. \tag{B.9}$$

Equation (B.7) is a heat-like equation, similar to the the Cole-Hopf transformation (Evans, 2022) for the Eikonal version of (HJ-Visc)[5].

It follows that the unique bounded fundamental solution of (B.8) is given by the Green's function convolution,

$$\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = \frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega \exp\left(-\frac{1}{2\delta t}\mid \boldsymbol{x}-\boldsymbol{y}\mid^2\right)\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)d\boldsymbol{y}, \;\; (\boldsymbol{x}\in\Omega, t>0) \tag{B.10a}$$

$$\triangleq \mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp\left(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{y})\right)\right] := \mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right] \tag{B.10b}$$

where (B.10a) is the standard heat kernel convolution and the kernel has been rewritten as a Gaussian density with mean $\boldsymbol{x}$ and variance δt. Note that $\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = 0$ elsewhere. □

Proof of Lemma 3.2. Going by equation (B.7), we can write

$$\boldsymbol{v}^\delta = -(\delta/2)\frac{|D\boldsymbol{v}^\delta|^2}{\boldsymbol{H}(t;\boldsymbol{x}, D\boldsymbol{v}^\delta)}\log\boldsymbol{\omega}^\delta := -\frac{1}{\boldsymbol{c}}\log\boldsymbol{\omega}^\delta, \tag{B.11}$$

so that the unique bounded solution to the initial-value (HJ-Visc) i.e. (12) becomes

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}\cdot\log\left\{\frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega \exp\left(-\frac{1}{2\delta t}\cdot|\boldsymbol{x}-\boldsymbol{y}|^2\right)\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right\}, \tag{B.12a}$$

$$\triangleq -\frac{1}{\boldsymbol{c}}\cdot\log\left\{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right]\right\}. \tag{B.12b}$$

Corollary B.2. *For the* (HJI-RCBRT-Visc) *and* (HJI-RCBRAT-Visc), $\boldsymbol{g}(\boldsymbol{y})$ *the above equation transforms to* $\boldsymbol{g}(\boldsymbol{x}+\sqrt{\delta(T-t)}\boldsymbol{y})$. *For numerical stability, we use the log-sum-exp identity*

$$\boldsymbol{v}^\delta(t;\boldsymbol{x}) = \frac{-1}{\boldsymbol{c}}\log\frac{1}{N}\sum_{i=1}^{N}\exp(-\boldsymbol{c}\boldsymbol{g}(\boldsymbol{x}+\sqrt{\delta(T-t)}\boldsymbol{y}_i)) \quad \textit{with} \;\; \boldsymbol{y}_i\sim\mathcal{N}(\boldsymbol{x},\delta t).$$

A fortiori, we have the solution to the viscous HJ equation as the log of the expectation of a Gaussian density with mean $\boldsymbol{x}$ and variance δt. □

B.2. Spatial Gradient of the HJ Payoff

Proof of Lemma 3.3. From (B.10a) observe,

$$D\boldsymbol{\omega}^\delta(t;\boldsymbol{x}) = -\frac{1}{(\sqrt{2\pi\delta t})^n}\int_\Omega \frac{(\boldsymbol{x}-\boldsymbol{y})}{\delta\cdot t}\exp\left(-\frac{\mid\boldsymbol{x}-\boldsymbol{y}\mid^2}{2\delta t}\right)\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)d\boldsymbol{y}, \tag{B.13a}$$

$$= -\frac{1}{\delta t}\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[(\boldsymbol{x}-\boldsymbol{y})\exp\left(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y})\right)\right]. \tag{B.13b}$$

Inspecting (B.11), we may write

$$D\boldsymbol{v}^\delta(t;\boldsymbol{x}) = -\frac{1}{\boldsymbol{c}}D[\log\boldsymbol{\omega}^\delta(t;\boldsymbol{x})] = -\frac{1}{\boldsymbol{c}}\frac{D\boldsymbol{\omega}^\delta(t;\boldsymbol{x})}{\boldsymbol{\omega}^\delta(t;\boldsymbol{x})} = \frac{1}{\delta\cdot t\cdot\boldsymbol{c}}\cdot\frac{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[(\boldsymbol{x}-\boldsymbol{y})\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}, \tag{B.14}$$

$$\triangleq \frac{1}{t\cdot\delta\cdot\boldsymbol{c}}\cdot\left(\boldsymbol{x}-\frac{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\boldsymbol{y}\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}{\mathbb{E}_{\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{x},\delta t)}\left[\exp(-\boldsymbol{c}\cdot\boldsymbol{g}(\boldsymbol{y}))\right]}\right). \tag{B.15}$$

□

[5] For the Eikonal version of (HJ-Visc), we set $\boldsymbol{H}(D\boldsymbol{v}(\boldsymbol{x})) = 0$ in Ω, and $\boldsymbol{v}(0;\boldsymbol{x}) = 0$ on $\partial\Omega$.

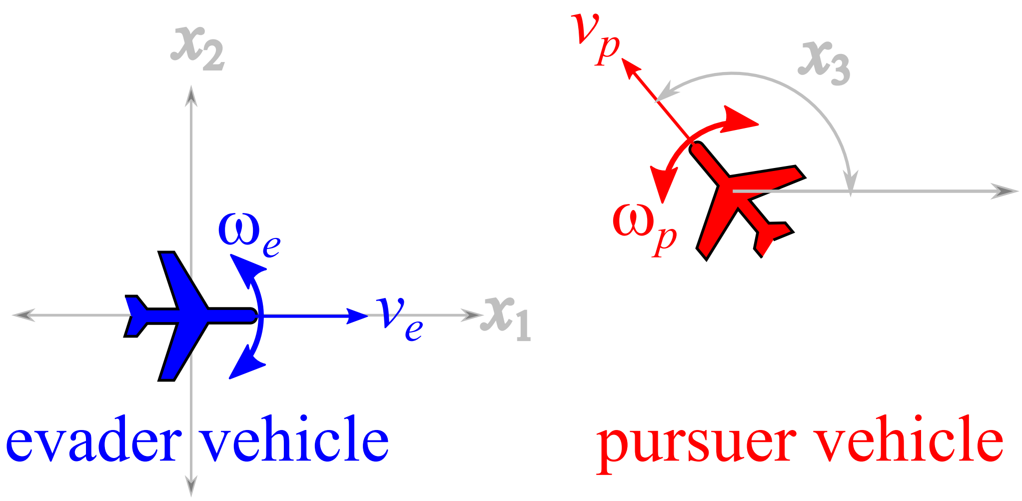

Figure 5. Two Dubins' vehicles in relative Cartesian coordinates. Reprinted from Mitchell (2020).

C. Further Numerical Results

This example was originally proposed by Merz (1972) as an iteration upon Isaacs (1999)'s homicidal chauffeur game, whereupon a pursuit-evasion game between two players with similar speeds and minimum turn radii, is thoroughly analyzed. In Mitchell (2020), this problem was established as a benchmark for testing the solubility of capturable set of states (the backward reachable tube) in Merz's classical pursuit-evasion game. In this example, we solve the problem with our LevelSetPy toolbox and establish that the approximated barrier surface to the two-player game conforms with standard results.

The game is that of two cars sharing similar Dubins dynamics (Dubins, 1957): $\boldsymbol{P}$ and $\boldsymbol{E}$ both have a positive minimum turn radii, w, and constant speeds v – with motion restricted to a plane as we have for the rocket launch differential game above. In relative coordinates, the diagrammatic structure of the motion is as depicted in Fig. 5. Choosing the Cartesian coordinate for motion representation, the state vector of the game with $\boldsymbol{E}$ at the origin can be characterized by its x_1, x_2 position relative to $\boldsymbol{P}$ and the angle θ between the two vehicles. Capture occurs when the distance $\|\boldsymbol{PE}\|_2$ between the pursuer and the evader becomes less than a specified radius.

The relative equations of motion, going by Fig. 5, is

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} -v_e + v_p \cos x_3 + w_e x_2 \\ v_p \sin x_3 - w_e x_1 \\ w_p - w_e \end{pmatrix}. \tag{B.16}$$

We adopt specialization to a case where the two vehicles only possess a unit velocity and unit maximum turn rates. Here, as Merz notes, if the initial velocities are parallel such as $x_3 = 0$, then the equations of relative motion imply that $\boldsymbol{E}$ can be separated from $\boldsymbol{P}$ forever by the initial radial separation if it replicates $\boldsymbol{P}$'s strategy. Whence, the barrier surface is closed and we are presented with Isaacs (1999)'s game of kind where we must determine the nature of the surface. This terminal surface possesses a closed-form solution and we refer readers to the treatment by Merz (1972). In this example, our chief concern is to judge the efficacy of our toolbox with respect to the analytical solution of the barrier surface.

The the backward reachable tube that consists of the paths taken by the trajectories of either player is defined as in the rockets pursuit evasion game so that we have

$$\ell(0, x) = \{x \in \mathcal{X} | x_1^2 + x_2^2 \leq r^2\}, \tag{B.17}$$

where again r is the capture radius. The target set is a cylinder as ℓ above excludes the heading, x_3. It is represented as shown in Fig. 6.

For a detailed treatment of the barrier surface, we refer readers to a proper analysis as elucidated in (Mitchell, 2020). Here, we focus on the construction of the BUP. The set of states that constitute the useable part and its boundary are respectively a

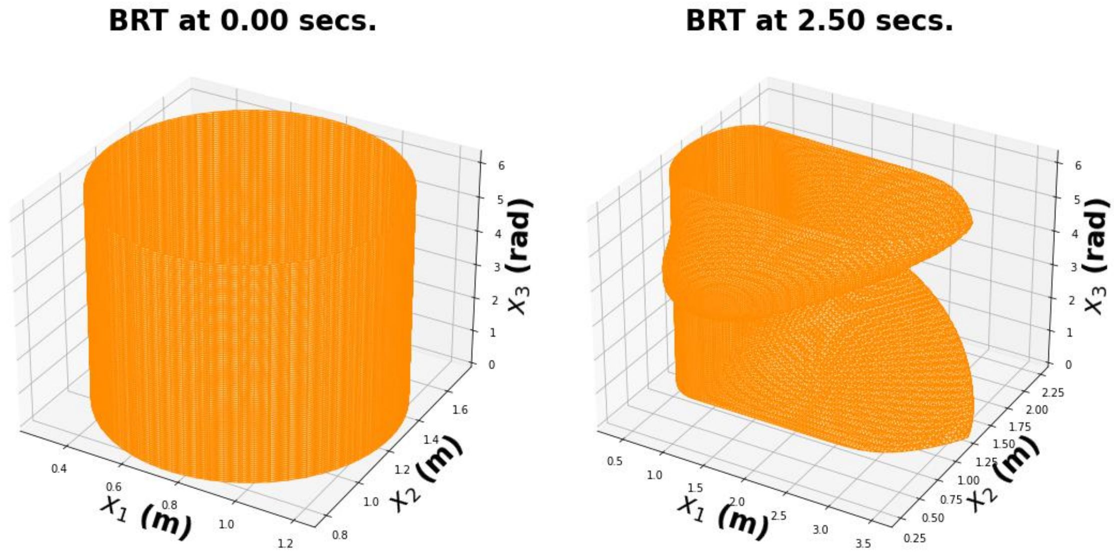

Figure 6. The target set (left) and the boundary of the useable part of the state space after the differential game between $\boldsymbol{P}$ and $\boldsymbol{E}$.

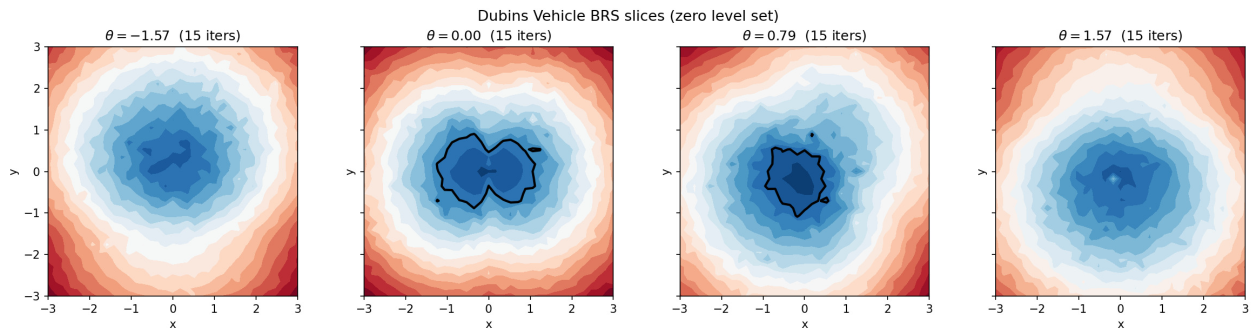

Figure 7. Dubins Vehicle BRT Slices

function of the implicit surface function representation $\ell : [-T, 0] \times \mathcal{X} \to \mathbb{R}$ so that for a $t \in [0, T]$, where $T > 0$ is

$$\mathcal{T} = \{x \in \mathcal{X} | \ell(0, x) \le 0\} \tag{B.18}$$

$$R([-t, 0], \mathcal{T}) = \{x \in \mathcal{X} | \ell(t, x) \le 0\}, \tag{B.19}$$

When $t > 0$, the implicit surface representation is the following HJI PDE

$$\frac{\partial}{\partial t}\ell(t, x) + \min\left(0, \boldsymbol{H}(x, \nabla_x \ell(t, x))\right) = 0. \tag{B.20}$$

It is easy to verify that the Hamiltonian is

$$H(x, p) = p_1(v_e - v_p \cos x_3) - p_2(v_p \sin x_3) - w|p_1 x_2 - p_2 x_1 - p3| + w|p_3|. \tag{B.21}$$

Since we are concerned with the special case that the linear and angular speeds are equal, we set $v_e = v_p = w \triangleq +1$ in the foregoing so that the Hamiltonian, in the final analysis is

$$H(x, p) = p_1(1 - \cos x_3) - p_2(\sin x_3) - |p_1 x_2 - p_2 x_1 - p3| + |p_3|. \tag{B.22}$$

The results and comparisons are provided in Fig. 7 and Fig. 8.

C.1. The Double Integral Plant

Here, we analyze a time-optimal control problem to determine what admissible control[6] can "transport" the system under consideration to a desired "origin" in the shortest possible time. We consider the double integral plant (Bhat & Bernstein, 1998; Wonham, 1985) as an illustrative example of our objective, which is to compute the points in the state space that can reach the origin in *finite-time* under the influence of a time-optimal controller.

We shall leverage standard necessary conditions from the principle of optimality (Bellman, 1957) to obtain a time-optimal feedback control design; introduce the notion of isochrones and switching surfaces; and discuss the analytic and approximate solutions (with our library) to the time-optimal control problem for a double integrator. We shall conclude the section by comparing the analytic and the overapproximated numerical solution (using the LevelSetPy toolbox) to the time to reach the origin problem.

C.1.1. Dynamics and Problem Setup

The double integrator is controllable, so that open-loop strategies may be employed in driving specific states to the origin in finite time (Wonham, 1985). The plant has the following second-order dynamics

$$\ddot{\boldsymbol{x}}(t) = \boldsymbol{u}(t) \tag{B.23}$$

and admits bounded control signals $\mid \boldsymbol{u}(t) \mid \le 1$ for all time t. After a change of variables,we have the following system of first-order differential equations

$$\begin{aligned} \dot{\boldsymbol{x}}_1(t) &= \boldsymbol{x}_2(t), \\ \dot{\boldsymbol{x}}_2(t) &= \boldsymbol{u}(t), \quad \mid \boldsymbol{u}(t) \mid \le 1. \end{aligned} \tag{B.24}$$

The *reachability problem that we consider is to address the question of what states can reach a certain point (here, the origin) in a transient manner*. That is, we would like to find point sets on the state space, at a particular time step, such that we can bring the system to the equilibrium, $(0, 0)$.

C.1.2. Time-optimal control scheme

This is an $\boldsymbol{H}$-minimal control problem whereupon we must find the control law that minimizes the Hamiltonian

$$\boldsymbol{H}(\boldsymbol{x}, p) = p_1 \dot{\boldsymbol{x}}_1 + p_2 \dot{\boldsymbol{x}}_2. \tag{B.25}$$

The necessary optimality condition stipulates that the minimizing control law be

$$\boldsymbol{u}(t) = -\text{ sign }(p_2(t)) \triangleq \pm 1. \tag{B.26}$$

For the co-states in question, suppose that their initial values (for constants k_1 and k_2) are $p_1(t_0) = k_1$ and $p_2(t_0) = k_2$, only four candidates can serve as time-optimal control sequences i.e. $\{[+1], [-1], [+1, -1], [-1, +1]\}$. On a finite time interval, $t \in [t_0, t_f]$, the time-optimal $\boldsymbol{u}(t)$ is a constant $k \equiv \pm 1$ so that for initial conditions $\boldsymbol{x}_1(t_0) = \boldsymbol{\xi}_1$ and $\boldsymbol{x}_2(t_0) = \boldsymbol{\xi}_2$, it can be verified that the state trajectories obey the relation

$$\boldsymbol{x}_1(t) = \boldsymbol{\xi}_1 + \frac{1}{2} k \left(\boldsymbol{x}_2^2 - \boldsymbol{\xi}_2^2\right), \text{ for } t = k\left(\boldsymbol{x}_2(t) - \boldsymbol{\xi}_2\right). \tag{B.27}$$

The trajectories of (B.27) traced out over a finite time horizon $t = [-1, 1]$ with *piecewise constant control laws*, $u = \pm 1$ on a state space and under the control laws $\boldsymbol{u}(t) = \pm 1$ is depicted in **??**. Curves with arrows that point upwards denote trajectories under the control law $\boldsymbol{u} = +1$; call these trajectories $\boldsymbol{\gamma}_+$; while the trajectories marked by dashed arrows pointing downward on the curves were executed under $\boldsymbol{u} = -1$; call these trajectories $\boldsymbol{\gamma}_-$.

[6] A control law is admissible when its range belongs in the admissible input set where it is bounded.

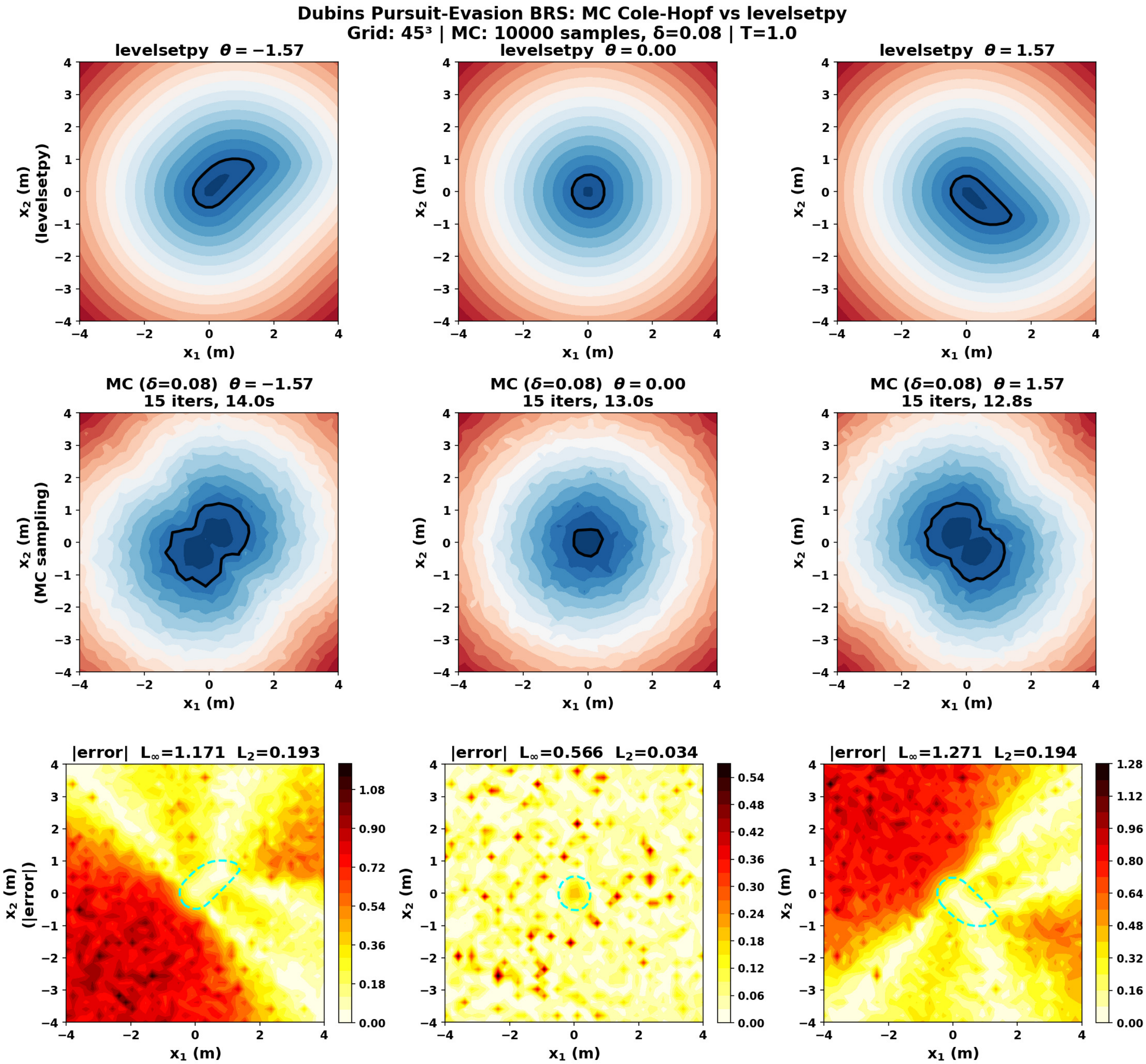

Figure 8. Dubins Vehicles' BRT Slices: Levelsetpy vs Monte-Carlo (Ours). The bottom row shows a heat map of the errors at various relative vehicle orientations (the control input) between the Monte Carlo sampling scheme and the levelsetpy scheme.

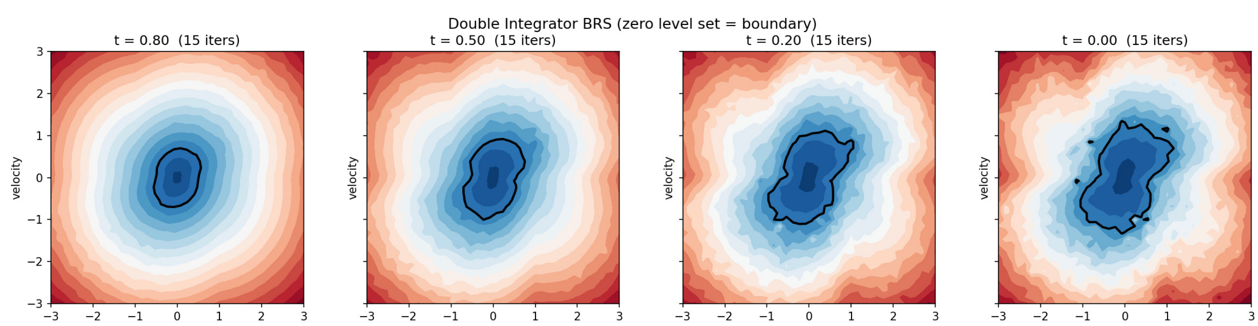

Figure 9. The time to reach the origin as 2D slice comparison for the double integratral plant